

Skills Summary

Basic Matrix Operations

Use this reference while completing your worksheet or when studying.

Skill 1 — Adding and subtracting matrices

Rule: same shape, entry by entry. The entry in row i , column j of $A + B$ is the entry of A there plus the entry of B there.

Shapes must match exactly, or the sum does not exist. Subtraction works the same way, one difference per position.

Example: Add the two tables of counts.

Step 1. Both matrices are 2×2 , so the sum exists and each position is settled on its own — no entry interacts with any other.

Step 2. $\begin{bmatrix} 3 & 5 \\ 7 & 2 \end{bmatrix} + \begin{bmatrix} 4 & 1 \\ 8 & 6 \end{bmatrix}$: add $3 + 4$, $5 + 1$, $7 + 8$, $2 + 6$.

$$\begin{bmatrix} 7 & 6 \\ 15 & 8 \end{bmatrix}$$

Try it: $\begin{bmatrix} 9 & 2 \\ 4 & 7 \end{bmatrix} + \begin{bmatrix} 3 & 6 \\ 5 & 1 \end{bmatrix}$

$\begin{bmatrix} 8 & 6 \\ 8 & 7 \end{bmatrix}$ check:

Skill 2 — Multiplying a matrix by a number

Rule: a scalar multiplies *every* entry once. The shape never changes.

In context: raising every price by 20% is the scalar 1.2; combining, as in $3A - 2B$, means scale each matrix first, then add or subtract entry by entry.

Example: Find $4 \begin{bmatrix} 2 & 5 \\ 3 & 1 \end{bmatrix}$.

Step 1. A scalar touches every position and nothing else changes, so the answer is the same shape with each entry multiplied once.

Step 2. $4(2)$, $4(5)$, $4(3)$, $4(1)$ — four separate products, no cross terms.

$$\begin{bmatrix} 8 & 20 \\ 12 & 4 \end{bmatrix}$$

Try it: $3 \begin{bmatrix} 4 & 1 \\ 2 & 6 \end{bmatrix}$

$\begin{bmatrix} 12 & 9 \\ 6 & 18 \end{bmatrix}$ check:

Skill 3 — Multiplying two matrices

Shape rule: an $m \times n$ times an $n \times p$ gives an $m \times p$. The two inner numbers must agree, or the product does not exist.

Entry rule: the entry in row i , column j pairs row i of the left matrix with column j of the right one, multiplying matched positions and adding the results.

Example: Find $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 5 & 1 \\ 2 & 6 \end{bmatrix}$.

Step 1. Inner dimensions agree (2 and 2), so the product exists and is 2×2 — four entries, each a row-times-column total.

Step 2. Row 1 with column 1: $2(5) + 3(2)$. Row 1 with column 2: $2(1) + 3(6)$.

Step 3. Row 2 with column 1: $1(5) + 4(2)$. Row 2 with column 2: $1(1) + 4(6)$.

$$\begin{bmatrix} 16 & 20 \\ 13 & 25 \end{bmatrix}$$

Try it: $\begin{bmatrix} 3 & 2 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 7 \end{bmatrix}$

$\begin{bmatrix} 27 \\ 26 \end{bmatrix}$:check

Skill 4 — A system as a matrix equation

The translation: coefficients go in the matrix, one row per equation and one column per unknown; the unknowns form a column; the right-hand sides form a column. That is $A\mathbf{x} = \mathbf{b}$.

Then solve as usual — substitution or elimination. The matrix form is what makes the structure visible, not a new solving method.

Example: Write $3x + 2y = 16$ and $x - y = 2$ as a matrix equation, and solve.

Step 1. Each equation becomes one row of coefficients, so the unknowns line up in columns and the constants form the right-hand column.

Step 2. $\begin{bmatrix} 3 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 16 \\ 2 \end{bmatrix}$

Step 3. From the second row $x = y + 2$; substituting, $3(y + 2) + 2y = 16$, so $5y = 10$.

$$x = 4, \quad y = 2$$

Try it: Write $2x + y = 11$ and $x + 3y = 18$ as a matrix equation, and solve.

$\mathbf{x} = \mathbf{b} \cdot \mathbf{g} = \mathbf{x}$:check